

Variables Guidelines

Systematic Variable Management for Modern Design Systems

System Architecture

- **01 Categories:** High-level taxonomy of variable types.
- **03 Static Variables:** Immutable foundations.
- **04 Dynamic Variables:** Adaptive logic and multi-mode systems.
- **05 Semantic Variables:** Functional and intent-based naming.
- **06 Non-semantic Variables:** Raw primitive foundations.



01 Variable Categories



Primitives

Global, unopinionated values (Colors, Spacing, Typography scales). Often referred to as Reference or Base tokens.



Semantic

Functional aliases that map primitives to specific roles within the UI (Backgrounds, Text, Borders).



Component

Highly specific tokens scoped to individual UI elements like Buttons, Inputs, or Navbars.

Token Mapping Flow

```
// The Chain of Design Decisions  
primitive: --blue-500: #007AFF;  
semantic: --bg-primary-action: var(--blue-500);  
component: --btn-cta-bg: var(--bg-primary-action);
```

Mapping allows us to change the **"What"** (Primitive) or the **"How"** (Semantic) without breaking the **"Where"** (Component).

03 Static Variables

Immutable Foundations

Static variables represent values that do not change regardless of the theme or context. They ensure visual consistency for branding and core geometry.

- Brand colors that must remain consistent (Logos).
- Layout grids and breakpoints.
- Standardized border-radius or line-heights.

```
:root {  
  --brand-name: "Horizon";  
  --grid-columns: 12;  
  --font-weight-bold: 700;  
}
```


04 Dynamic Variables

 Code Editor with Variables

Context-Aware Logic

Dynamic variables support **Modes**. These values change automatically based on user settings or system environment (e.g., Light Mode, Dark Mode, High Contrast).

Instead of hardcoding colors, we use logic-based switching to maintain accessibility and user preference.

System Evolution

Feature	Static	Dynamic
Reliability	Consistent across all themes	Adapts to current mode
Use Case	Core brand, Spacing, Typography	Color backgrounds, Text roles
Implementation	Single value definition	Multiple values (Modes)

05 Semantic

Naming by Intent, not Appearance.

Semantic variables provide meaning to your design.
Instead of `blue-500`, we use `action-primary`.

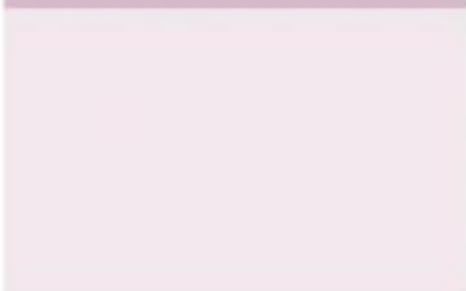
- `--text-error` (not red)
 - `--surface-main` (not white)
 - `--border-subtle` (not grey-10)
-

06 Non-semantic

Primitive Foundations

Non-semantic variables (Primitives) are the raw palette of your system. They contain the hex values, pixel counts, and specific measurements without assigning meaning.

Note: Primitives should rarely be used directly in components. They serve as the "Source" for semantic aliases.

	-2		-3	
	-1		-2	
	PURPLE		-1	
	+1		TURQUOISE	
	+2		+1	
	+3		+2	
	+4		+3	

System Impact

40%

Reduction in CSS Payload

Scalability Wins

- Faster handoff between design and engineering.
- Global brand updates take seconds, not weeks.
- Ensured WCAG 2.1 Contrast compliance across themes.

Naming Conventions



Avoid Names

Never use color names (Red, Blue) or positions (Top, Left) in semantic tokens.



Taxonomy

Use [Category] - [Role] - [State] logic.

Ex: `bg-primary-hover`



Kebab-case

Standardize on `kebab-case` for code-friendly handoff to CSS/SASS.

Q&A

Thank you for exploring Variable Guidelines.

Horizon Design System Team

Image Sources



<https://static.vecteezy.com/system/resources/thumbnails/069/274/178/small/abstract-network-visualization-with-glowing-nodes-and-connecting-lines-on-a-dark-purple-background-futuristic-digital-data-or-neural-network-concept-ideal-for-tech-and-science-themes-video.jpg>

Source: www.vecteezy.com



Thumbnail for

github.com

<https://private-user-images.githubusercontent.com/18739364/580310752-c679bd35-ba7e-4d6f-834d-3f4b6e5a35e4.png?jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWluY29tliwiYXVkIjoicmF3LmdpdGh1YnVzZXJjb250ZW50LmNvbSlmltleSI6ImtleTUiLCJleHAiOjE3Nzc2Nzc5OTYslm5iZil6MTc3NzY3NzY5NiwicGF0aCI6Ii8xODczOTM2NC81ODAzMTA3NTItYyZ3OWJkMzUtYmE3ZS00ZDZmLTgzNGQtM2Y0YjZINWEzNWU0LnBuZz9YLUFtei1BbGdvcml0aG09QVdTNClITUFDLVNIQTl1NiZYLUFtei1DcmVkJW50aWFsPUFLSUFWQ09EWUxTQTUzUFFLNFPjBTJTGMjAyNjA1MDEIMkZ1cy1lYXN0LTEIMkZzMyUyRmF3czRfcmlvdCZYLUFtei1EYXRIPTIwMjYwNAxVDIzMDZmWC1BbXotRXhwaXJlc30zMDAmWC1BbXotU2lnbmF0dXJlPTJIODIlMmQyY2I0NTZkNzAxYzJiMTY0NWU5Zjc1NWQyYmRmMTY2ZTZqZmU3NjQxOWQ5ZTIiIN2QzODIzZDZkZjgmWC1BbXotU2lnbmVkcGVhZGVyc30zN0JnJlc3BvbnNILWNvb3RibnQtdHlwZT1pbWFnZSUyYSZrxdYSI209xeetW0WxeHD>

Source: github.com



Thumbnail

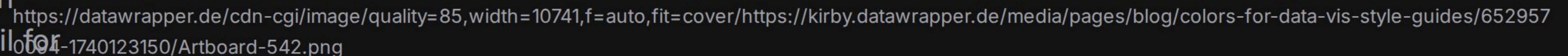
<https://duck.design/wp-content/uploads/2025/07/Split-Screen-Interface-Scenarios-Dark-and-Light-Modes-comparison-scaled.jpg>

Source: duck.design



Thumbnail for <https://www.youtube.com/watch?v=0094>

www.datawrappers.com



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